12TH BIOLOGY · CHAPTER 15 · FULL CHAPTER NOTES

Human Nervous System

Topic-wise short and long questions with answers from the complete chapter

15.1 NERVOUS COORDINATION

Q.1 [Short] What is coordination? Which two systems carry it out in animals?

Coordination means the working together of different parts of the body in response to stimuli. In animals it is carried out by the **nervous system** (nervous coordination) and the **endocrine system** (chemical coordination).

Q.2 [Long] Describe the three basic elements of nervous coordination.

1. **Receptors:** These are organs, tissues or cells, such as eyes, ears or skin, that detect stimuli (changes in the external and internal environment). They act as transducers. When they receive stimuli, they generate nerve impulses in sensory neurons, which carry the impulses to the central nervous system.
2. **Central nervous system (CNS):** It is the control centre of nervous coordination. It consists of the brain and spinal cord, which are made of associative neurons (inter-neurons). They receive nerve impulses from sensory neurons, process them and send impulses to effectors via motor neurons.
3. **Effectors:** These are the parts which produce responses on receiving nerve impulses from the CNS via motor neurons. Muscles and glands act as effectors. Muscles respond by contracting and glands by producing secretions.

Q.3 [Short] Why are receptors called transducers?

A transducer changes one form of energy into another. Receptors receive stimuli (light, sound, chemicals, pressure, etc.) and change them into nerve impulses in sensory neurons. That is why they act as transducers.

Q.4 [Short] State the path of a message transmitted through the nervous system.

Stimulus → Receptor → Sensory neuron → Central nervous system (inter-neurons) → Motor neuron → Effector (muscle or gland) → Response.

Q.5 [Short] How do muscles and glands respond as effectors?

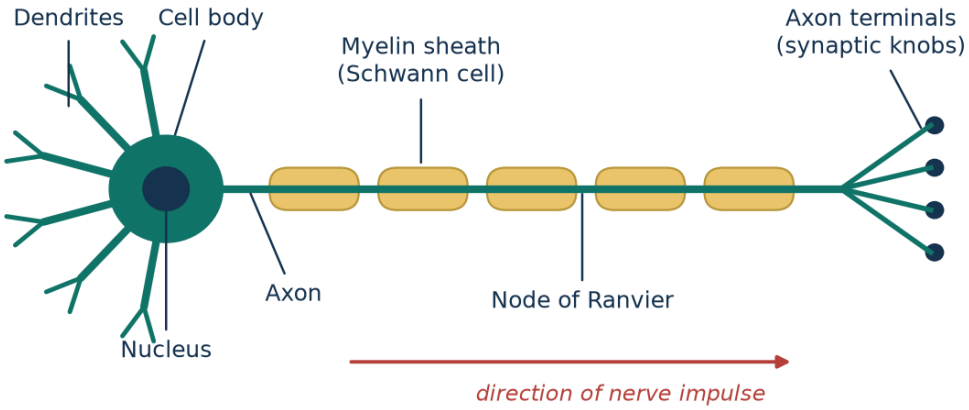
Muscles respond by contracting, and glands respond by producing secretions, when they receive nerve impulses from the CNS through motor neurons.

15.2 NEURONS

Q.6 [Long] Explain the structure of a neuron.

A neuron (nerve cell) is the structural and functional unit of the nervous system. There are many variations in the structure of neurons, but all neurons have three basic components: a cell body, dendrites and an axon.

- **Cell body:** It is an enlarged region of the neuron. The nucleus and most of the organelles are located in it. Its cytoplasm contains **Nissl's granules**, which are groups of ribosomes and rough endoplasmic reticulum.
- **Dendrites:** These are membrane-bounded extensions of the cell body. They receive information from other neurons or other cells, convert it into nerve impulses and carry it toward the cell body. Motor neurons and inter-neurons receive impulses from many sources at the same time, so they have highly branched dendrites.
- **Axon:** It is a long, membrane-bounded extension of the cell body. It transmits impulses away from the cell body. A neuron may have a single axon or branching axons. The end of an axon is called the **axon terminal**. It may contact and communicate with a muscle cell, a gland cell or another neuron.



Structure of a myelinated neuron (schematic)

Q.7 [Short] What are Nissl's granules?

Nissl's granules are groups of ribosomes and rough endoplasmic reticulum present in the cytoplasm of the cell body of a neuron.

Q.8 [Short] What are neuroglia? How is the myelin sheath formed and what is its function?

The special cells which support neurons are called **neuroglia** or glial cells. They supply nutrients to neurons, remove wastes and provide immunity. Two important glial cells are Schwann cells and oligodendrocytes. They wrap around the axons of many neurons and make **myelin sheaths** (sheaths of multiple membrane layers). Schwann cells make myelin sheaths in the peripheral nervous system, and oligodendrocytes make them in the central nervous system.

The myelin sheath is an insulating covering. It helps in the fast transmission of nerve impulses along the axon. It is interrupted at intervals of 1 to 2 mm by small non-myelinated gaps called **nodes of Ranvier**.

Q.9 [Short] Differentiate between Schwann cells and oligodendrocytes.

Both are glial cells which wrap around axons and make myelin sheaths. Schwann cells make the myelin sheaths of neurons in the peripheral nervous system, while oligodendrocytes make them in the central nervous system.

Q.10 [Long] Describe the three types of neurons on the basis of their functions.

Feature	Sensory neuron	Inter-neuron (associative)	Motor neuron
Function	Conducts impulses from receptors to the CNS	Receives impulses from sensory neurons or other inter-neurons and transmits them to motor neurons or other inter-neurons	Conducts impulses away from the CNS to effectors
Location	Cell bodies lie outside the CNS in clusters called ganglia	Occurs entirely within the CNS	Cell bodies lie in the CNS
Structure	Unipolar: one long dendrite on one side of the cell body and an axon on the other	Some have short axons (circuits with nearby neurons); others have long axons (connect different regions of brain and spinal cord)	Multipolar: cell body at one end, many branched dendrites and a long axon

The textbook describes sensory neurons as unipolar with "only one long dendrite".

Q.11 [Short] Why do motor neurons and inter-neurons have highly branched dendrites?

Motor neurons and inter-neurons receive impulses from many sources at the same time. Highly branched dendrites give them a large surface to receive all these impulses.

Q.12 [Long] What is a reflex arc? Explain reflex action with the example of the knee-jerk reflex.

In the nervous system, neurons follow specific neural pathways (arcs) to receive, transmit and process impulses. The **reflex arc** is the simplest neural pathway. It controls **reflex action**, i.e., an immediate, automatic and involuntary response to an external or internal stimulus.

Example: the knee-jerk reflex

1. When the tendon below the knee (the patellar tendon) is hit with a hammer, the tapping acts as a **stimulus**.
2. The stimulus is sensed by a **receptor** present in the quadriceps muscle of the thigh.
3. The receptor generates a nerve impulse in a **sensory neuron**, which carries the impulse to the spinal cord.
4. In the spinal cord, the sensory neuron transfers the impulse to a **motor neuron** without involving an inter-neuron.
5. The motor neuron carries the impulse to the **effector** (quadriceps muscle), which contracts. The lower leg extends at the knee, i.e., it kicks forward.
6. In the meantime, the sensory neuron also activates an **inter-neuron** in the spinal cord. The inter-neuron releases chemicals which inhibit the motor neurons of the hamstring muscles. So the hamstrings relax during the extension of the lower leg.

Q.13 [Short] Why do the hamstring muscles relax during the knee jerk?

The sensory neuron also activates an inter-neuron in the spinal cord. This inter-neuron releases chemicals which inhibit the motor neurons of the hamstring muscles. So the hamstrings relax and allow the quadriceps to extend the lower leg.

15.3 NERVE IMPULSE

Q.14 [Short] Define nerve impulse and membrane potential.

A **nerve impulse** is a wave of electrochemical changes that travels along the length of a neuron.

The inner side of a cell membrane is electrically negative while the outer side is positive. This difference in the electrical charges across the cell membrane is called the **membrane potential**.

Q.15 [Short] Why can the membrane potentials of neurons and muscle cells change, but not those of most other cells?

In most cells, the inward and outward movement of ions through the membrane is constant, so the net negative charge inside remains constant. The membranes of neurons and muscle cells can change the rates of ion movements, so their membrane potentials also change.

Q.16 [Long] What is resting membrane potential? How is it established?

A resting neuron (not conducting a signal) is said to be **polarized**. Its inner side has a negative charge while its outer side has a positive charge. This membrane potential of a polarized neuron is called the **resting membrane potential (RMP)**. It is about –70 mV. The negative sign shows that the inside of the cell is negative with respect to the outside. RMP is established by the following factors:

1. **Fixed anions:** Large negatively charged molecules, e.g., proteins and nucleic acids, are more abundant inside the cell and cannot diffuse out.
2. **Sodium–potassium pump:** The concentration of K^+ is 30 times greater inside the neuron than outside, while the concentration of Na^+ is 10 times greater outside than inside. Sodium–potassium pumps constantly move these ions against their concentration gradients by using ATP. For every two K^+ transported inward, three Na^+ are pumped out. So the inside becomes more negative than the outside.
3. **Leakage of K^+:** Some K^+ ions also leak out of the cell through the K^+ gates of the membrane.

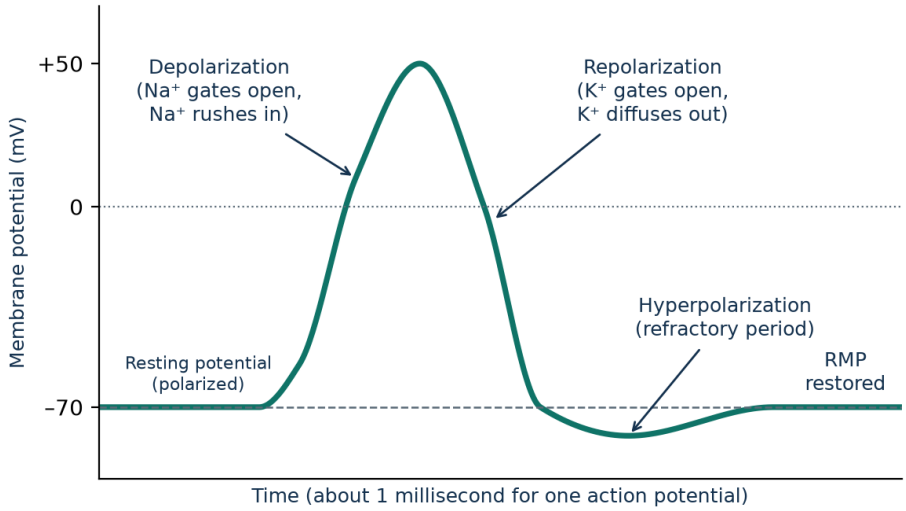
Q.17 [Short] What does the negative sign of –70 mV indicate?

The negative sign indicates that the inside of the neuron is negative with respect to the outside.

Q.18 [Long] What is an action potential? Describe the events of depolarization, repolarization and hyperpolarization.

When a neuron becomes active (generates a nerve impulse), the membrane potential at a region of the neuron is reversed, i.e., the inside becomes positive with respect to the outside. It is called the **active membrane potential** or **action potential**. It is produced by the following events:

- **Depolarization:** The Na^+ gates of the membrane open and Na^+ ions rush into the cell. This reduces the negative charge on the inside. The membrane potential changes from –70 mV towards zero and then reaches +50 mV. This reversal of polarity is called depolarization. It produces an action potential on a short region of the neuron for about one millisecond.
- **Repolarization:** After a short period, the Na^+ gates close and the K^+ gates open. More K^+ ions diffuse out. This makes the inside of the cell negative again (negative inside and positive outside).
- **Hyperpolarization:** Repolarization carries the membrane potential to a value slightly more negative than the resting potential for a brief period. It is due to a slight delay in the closing of all K^+ gates. During hyperpolarization, the neuron cannot generate another action potential. This period is called the **refractory period**.
- **Return to RMP:** After the short refractory period, the original distribution of ions is restored by the sodium–potassium pump and the membrane returns to its RMP, i.e., –70 mV. Now this region is ready for the next action potential.



Graph of an action potential

Q.19 [Short] What is the refractory period? What causes it?

The refractory period is the brief period of hyperpolarization during which the neuron cannot generate another action potential. It is caused by a slight delay in the closing of all K^+ gates, which makes the membrane slightly more negative than the resting potential.

Q.20 [Short] How does a nerve impulse move along the axon?

In reality, action potentials (impulses) do not travel along the axon. Rather, they are reproduced at adjacent regions along the axon. An action potential in one region of the axon serves as a depolarization stimulus for the next region. So the next region produces its own action potential. Meanwhile, the previous region repolarizes back to the RMP.

Q.21 [Short] Which factors affect the velocity of a nerve impulse?

The strength (amplitude) of action potentials remains constant along the length of the axon. However, the velocity of the nerve impulse may vary. The velocity is greater if the diameter of the axon is large or if the axon is myelinated.

Q.22 [Long] Describe the saltatory conduction of nerve impulse.

In myelinated axons, action potentials are produced only at the **nodes of Ranvier**. The action potential jumps from one node of Ranvier to the next, skipping the myelinated regions of the membrane. This is called **saltatory conduction** of the nerve impulse.

Saltatory conduction is up to 50 times faster than the conduction of a nerve impulse through the fastest non-myelinated axons. In humans, non-myelinated neurons conduct impulses at a speed of 1 to 3 metres per second, while myelinated neurons conduct at speeds of up to 120 metres per second.

Q.23 [Short] Why is conduction faster in myelinated neurons?

Myelin is an insulating covering, so action potentials are produced only at the nodes of Ranvier. The impulse jumps from node to node and skips the myelinated regions. So it covers the same distance in much less time (up to 120 m/s compared with 1 to 3 m/s).

15.4 SYNAPSE

Q.24 [Long] Describe the structure of a synapse.

The junction between the axon terminal of one neuron and the dendrites of another neuron, or an effector cell (of a muscle or gland), is called a **synapse**. Its parts are:

- **Presynaptic neuron:** The neuron whose axon transmits action potentials towards the synapse. Its axon terminal has several **synaptic knobs**. These knobs contain numerous **synaptic vesicles**, which are packed with chemicals called neurotransmitters.
- **Synaptic cleft:** At a synapse, the two cells are not in direct contact. There is a fluid-filled gap between them, called the synaptic cleft.
- **Postsynaptic cell:** The neuron or cell on the other side of the synapse. Its membrane has receptor proteins for neurotransmitters.

The synapse between a motor neuron and a muscle fibre is also called a **neuromuscular junction**.

Q.25 [Long] Explain the synaptic transmission of a nerve impulse.

The movement of an action potential across a synapse is called **synaptic transmission**. It happens in the following steps:

1. When an action potential arrives at the synaptic knobs, calcium gates in the presynaptic membrane open.
2. Calcium ions present in the synaptic cleft diffuse rapidly into the presynaptic axon.
3. Due to the entry of calcium ions, the synaptic vesicles fuse with the presynaptic membrane and release neurotransmitters by exocytosis.
4. The released neurotransmitters diffuse rapidly across the cleft and bind to receptor proteins on the postsynaptic membrane.
5. This binding acts as a message for the postsynaptic membrane. Its Na^+ gates open and an action potential is generated in the postsynaptic membrane.
6. After their action, the neurotransmitters do not remain in the cleft. At many synapses, they are reabsorbed by the presynaptic neuron. At other synapses, they are broken down by enzymes.

Q.26 [Short] What is the role of calcium ions in synaptic transmission?

When an action potential reaches the synaptic knob, calcium gates open and calcium ions diffuse into the presynaptic axon. They cause the synaptic vesicles to fuse with the presynaptic membrane and release neurotransmitters by exocytosis.

Q.27 [Short] What happens to neurotransmitters after their action?

They do not remain in the synaptic cleft. At many synapses they are reabsorbed by the presynaptic neuron. At other synapses they are broken down by enzymes.

Q.28 [Short] What is a neuromuscular junction?

The synapse between a motor neuron and a muscle fibre is called a neuromuscular junction.

Q.29 [Long] What are neurotransmitters? Describe excitatory neurotransmitters with examples.

Neurotransmitters are chemical messengers which neurons use to communicate with each other and with their target cells. They are excitatory or inhibitory.

Excitatory neurotransmitters generate a nerve impulse (action potential) in postsynaptic neurons by increasing the permeability of the membrane to Na^+. Examples:

- **Acetylcholine:** acts between motor neurons and skeletal muscle fibres, and between many neurons.
- **Glutamate:** acts in the CNS.
- **Biogenic amines** (they also function as hormones): **norepinephrine** (made in the CNS and sympathetic nerves; regulates sleep patterns, focus and alertness), **epinephrine** (released in adrenal glands and brainstem; plays a role in sleep and alertness), **dopamine** (in the brain; controls body movements, pleasure related to motivation and emotional stimulation) and **serotonin** (about 90% produced in the intestine and 10% in the CNS; regulates appetite, sleep, memory and learning,

temperature, mood and behaviour).

Q.30 [Long] How do inhibitory neurotransmitters work? Give examples.

Inhibitory neurotransmitters decrease the chance of transmission of a nerve impulse to the postsynaptic neuron. When they bind to the receptors on the postsynaptic neuron, channels for chloride ions open instead of sodium channels. Chloride ions diffuse into the neuron and make the inside of the membrane even more negative than the RMP. This hyperpolarization causes the inhibitory effect. Examples:

- **GABA (gamma-aminobutyric acid):** the chief inhibitory neurotransmitter of the CNS. It reduces neural excitability, reduces anxiety, stress and fear, and promotes relaxation and sleep.
- **Glycine:** an inhibitory neurotransmitter, especially in the spinal cord, brainstem and retina.
- **Endorphins:** released by neurons in the brain. They reduce the release of the excitatory neurotransmitters that transmit pain messages, so they block pain signals.

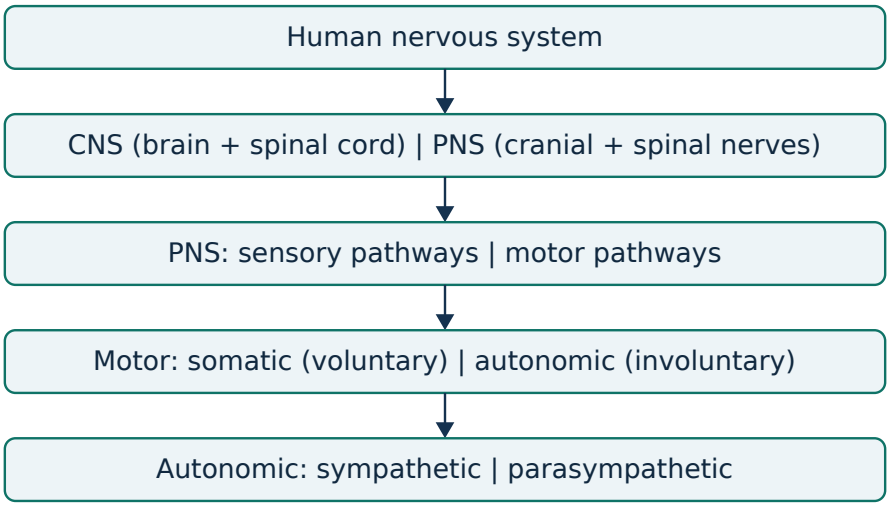
Q.31 [Short] How do endorphins reduce pain?

Endorphins are released by neurons in the brain. They reduce the release of the excitatory neurotransmitters that transmit pain messages. So they block pain signals. Opium and its derivatives (morphine and heroin) reduce pain because they are similar in chemical structure to endorphins.

15.5 BASIC ORGANIZATION OF HUMAN NERVOUS SYSTEM

studysapiens.com

Q.32 [Short] Give the basic organization of the human nervous system.



Organization of the human nervous system

Q.33 [Short] What are meninges? Name and describe their layers.

The brain is enclosed within the cranium and the spinal cord within the vertebral column. Both are covered with protective **meninges**, which consist of three layers of membranes:

1. **Dura mater:** the outermost thick layer, below the skull and vertebrae. It contains larger blood vessels.
2. **Arachnoid:** the thin second layer. It acts as a barrier between the cerebrospinal fluid present below it and the blood in the dura mater.
3. **Pia mater:** the third, very delicate layer. It adheres to the surfaces of the brain and spinal cord and has blood capillaries that nourish them.

Q.34 [Long] Describe the structure and functions of the cerebrum.

It is the largest portion of the human brain. Its outer layer is highly folded. A deep groove (fissure) divides it into the left and right **cerebral hemispheres**, which are connected at the base of the groove by the **corpus callosum** (a band of axons). Each hemisphere receives sensory information from the opposite side of the body and sends motor commands to that side. Each hemisphere has four lobes:

- **Frontal lobes:** executive functions including self-control, planning, reasoning and abstract thought.
- **Parietal lobes:** process sensory information, give awareness of the position of the body and the environment, control attention, and plan and execute motor movements.
- **Temporal lobes:** process sensory input for retention of visual memory, language comprehension and emotions.
- **Occipital lobes:** control vision.

The **cerebral cortex** is the folded outer layer of the cerebrum, made of grey matter. It contains **motor areas** (send impulses to voluntary skeletal muscles), **sensory areas** (receive information from different parts, e.g., the auditory cortex in the temporal lobe

and the visual cortex in the occipital lobe) and **association areas** (sites of higher mental activities). Below the cortex is white matter, which links regions of the cortex with each other and with other neural centres.

Q.35 [Short] What is the corpus callosum?

It is a band of axons at the base of the deep groove of the cerebrum. It connects the left and right cerebral hemispheres.

Q.36 [Short] Describe the parts and functions of the brainstem.

The cerebrum is connected to the spinal cord through the brainstem. It lies below the cerebrum and has three divisions:

- **Midbrain:** relays visual and auditory information. It also contains the reticular formation, which connects the hindbrain with the forebrain.
- **Pons:** transmits impulses between the cerebral hemispheres and the cerebellum.
- **Medulla oblongata:** serves as both a relay centre and a control centre for heart rate, respiration rate and other homeostatic activities.

Q.37 [Short] Describe the diencephalon and the limbic system.

It is present deep within the centre of the brain, above the brainstem and beneath the cerebrum. It has two major parts:

- **Thalamus:** a relay centre between sensory information and the cerebrum.
- **Hypothalamus:** integrates the visceral activities. It regulates body temperature, hunger and satiety, thirst and various emotional states. It also controls the pituitary gland, which in turn regulates many other endocrine glands.

The **limbic system** is a group of linked structures on both sides of the thalamus, just under the cerebrum. It includes the hypothalamus, the amygdala and the hippocampus. The **amygdalae** control feelings and emotions of love, hate, anger, fear, rage and sexual arousal. The **hippocampus** plays a role in the consolidation of information from short-term memory to long-term memory.

Q.38 [Short] Write the location and functions of the cerebellum.

It is the second major structure of the brain. It lies below and behind the cerebral hemispheres. It consists of a tightly folded layer of cortex with white matter underneath. It receives sensory impulses from muscles, tendons, joints, eyes and ears, and information from other brain centres. It processes information about body position and controls posture. It coordinates muscle actions, especially rapid and ongoing movements, by acting with the brainstem and the cerebral cortex.

Q.39 [Short] What are ventricles of the brain?

The brain and spinal cord are hollow structures. They contain fluid-filled cavities called **ventricles**. There are four ventricles in the human brain: two large lateral ventricles in the cerebrum, the third ventricle in the diencephalon between the right and left thalamus, and the fourth ventricle between the cerebellum and the brainstem. The fourth ventricle is continuous with the central canal of the spinal cord. The main function of the ventricles is the production and circulation of cerebrospinal fluid.

Q.40 [Long] Describe the structure and function of the spinal cord.

The spinal cord is a cable of neurons extending from the medulla oblongata of the brain down through the backbone. It is enclosed and protected by the vertebral column and the meninges. There are two zones inside it:

- **Inner zone (grey matter):** It is H-shaped or butterfly-shaped. It contains the cell bodies of inter-neurons and motor neurons, neuroglia and non-myelinated axons. It also contains a central fluid-filled canal, which is a continuation of the fourth ventricle of the brain.
- **Outer zone (white matter):** It contains myelinated axons.

The spinal cord acts as an information highway on which messages are transmitted between the body and the brain.

Q.41 [Short] How is the arrangement of grey and white matter in the spinal cord different from that in the brain?

The arrangement is opposite. In the brain (cerebrum), grey matter is on the outer side (cortex) and white matter is below it. In the spinal cord, grey matter makes the inner H-shaped zone and white matter makes the outer zone.

Q.42 [Short] What are nerves and ganglia?

Nerves are cable-like collections of axons. Ganglia are aggregations of cell bodies of neurons located outside the CNS. The peripheral nervous system consists of nerves and ganglia.

Q.43 [Long] Write a note on cranial nerves.

The nerves which arise from the brain are called cranial nerves. There are **12 pairs** of cranial nerves. The first two pairs (olfactory and optic) arise from the cerebrum. The other 10 pairs arise from the brainstem. Some cranial nerves are sensory, some are motor and others are mixed (contain axons of both sensory and motor neurons).

Nerve	Nature	Major function
I	Sensory	Sense of smell
II	Sensory	Sense of vision
III	Motor	Movement of eye
IV	Motor	Movement of the superior oblique muscle of eye
V	Mixed	Sensation in the face; control of the muscles of chewing
VI	Motor	Movement of the lateral muscles of eye (outward gaze)
VII	Mixed	Facial expressions; sense of taste in the front of the tongue
VIII	Sensory	Hearing and balance
IX	Mixed	Sense of taste in the back of tongue; muscles of swallowing
X	Mixed	Controls many organs, including heart, lungs and digestive system

Nerve	Nature	Major function
XI	Motor	Controls the muscles of the neck and shoulders
XII	Motor	Controls the muscles of the tongue

Q.44 [Short] Write a note on spinal nerves.

The nerves which arise from the spinal cord are called spinal nerves. There are **31 pairs** of spinal nerves and all of them are mixed. They transmit sensory and motor information between the spinal cord and the rest of the body. Each spinal nerve emerges from the spinal cord by two short roots. The **dorsal root** contains the axons and dendrites of sensory neurons, whose cell bodies lie outside the spinal cord in the dorsal root ganglia. The **ventral root** contains the axons of motor neurons. The two roots join just before the nerve leaves the vertebral column.

The 31 pairs are: 8 pairs of cervical, 12 pairs of thoracic, 5 pairs of lumbar, 5 pairs of sacral and 1 pair of coccygeal nerves.

The textbook sentence reads "those which arise from spinal nerves are called spinal nerves". It should be "arise from the spinal cord".

Q.45 [Short] Differentiate between the sensory and motor pathways of the PNS.

Sensory pathways contain the sensory neurons of spinal and cranial nerves, which connect the receptors of the body to the CNS. Motor pathways contain the motor neurons of spinal and cranial nerves, which connect the CNS to the effectors of the body.

Q.46 [Short] What is the somatic nervous system?

It consists of those motor neurons of spinal and cranial nerves which control skeletal muscles. It is voluntary, i.e., skeletal muscles can be moved at will. For reflex actions, it operates without conscious control.

Q.47 [Long] Explain the types and functions of the autonomic nervous system.

The autonomic nervous system consists of those motor neurons of spinal and cranial nerves which control smooth muscles and glands. It is involuntary. It controls internal body conditions, e.g., respiration, heartbeat and digestion. It has two subdivisions.

(a) Sympathetic nervous system

It prepares the body under physical or emotional stress. Its motor neurons originate from the thoracic and lumbar regions of the spinal cord and reach different visceral organs (smooth muscles and glands). When it is activated, it prepares the body for **fight or flight**. For example, the heart beats faster and stronger, blood glucose concentration increases, blood flow is diverted to the muscles and heart, and the bronchioles dilate.

(b) Parasympathetic nervous system

It antagonises the response of the sympathetic system after the threat or stress has passed. Its motor neurons originate from the brain and the sacral region of the spinal cord. Many of them are in the vagus (tenth cranial) nerve. They bring the internal organs back to their normal conditions. For example, the heartbeat slows down, blood flow to the heart and skeletal muscles decreases, and the digestive organs increase their secretion and other activities.

Body part	Effect of sympathetic system	Effect of parasympathetic system
Pupil of eye	Dilation	Constriction
Salivary glands	Decreased secretion	Increased secretion
Gastric glands	Inhibition of secretion	Stimulation of secretion
Liver	Stimulation of glucose secretion	Inhibition of glucose secretion
Digestive tract	Decreased motility	Increased motility
Urinary bladder	Relaxation	Contraction
Heart muscle	Increased rate and strength	Decreased rate
Bronchioles	Dilation	Constriction
Blood vessels in muscles	Dilation	No effect
Blood vessels in skin	Constriction	No effect

15.6 RECEPTORS IN HUMAN BODY

Q.48 [Long] Explain the structure and functioning of the receptors for taste.

Structure

The sense of taste is detected by special chemoreceptors called **taste buds**. There are about 10,000 taste buds. Most of them are embedded between bumps called **papillae** on the tongue. Some are present in the throat and on the roof of the oral cavity. A taste bud is bulb-shaped. It is made up of receptor cells with hair-like structures and basal cells (for making new receptor cells). It opens into the mouth through a **taste pore**.

Functioning

When the chemoreceptor cells of taste buds are stimulated by the chemicals of food or drink, they produce nerve impulses in the associated sensory neurons. The impulses travel along cranial nerves VII, IX and X to the brainstem, and from there to the taste-sensing areas of the cerebral cortex. The cortex processes this information with other sensory information, such as smell and texture, to create the complete perception of taste.

Humans have four kinds of taste buds: salty, sweet, sour and bitter. They are concentrated in specific regions of the tongue: sweet at the tip, sour at the sides, bitter

at the back, and salty over most of the surface. The four regions overlap at certain places.

Q.49 [Long] Write a detailed note on the receptors for smell.

Structure

The chemoreceptors for smell (olfaction) are located in the upper part of the nasal cavities. These receptors are actually the ends of sensory neurons. The sensory neurons are present in a thin layer of tissue called the **nasal epithelium**. They are packed tightly between two other types of cells: **supporting cells**, which provide protection, and **basal cells**, which act as stem cells to replace old neurons. The dendrites of these sensory neurons have tiny hair-like structures called **cilia**, which project into the moist mucus lining the nasal cavity.

Functioning

When air enters the nasal cavities, the air-borne chemicals dissolve in the mucus. The dissolved chemicals bind to specific receptor sites on the cilia. This binding creates a nerve impulse in the sensory neurons. The impulse is transmitted to the **olfactory bulb** through the olfactory nerves (cranial nerve I). The olfactory bulb produces the perception of smell.

Q.50 [Short] What are nociceptors?

Nociceptors detect pain. Pain is defined as a stimulus that causes or is about to cause tissue damage. Nociceptors consist of free nerve endings located throughout the body, especially near surfaces. Different nociceptors may respond to extremes of temperature, very intense pressure or specific chemicals (including some released by injured cells).

Q.51 [Long] Describe the touch receptors of the skin.

The skin contains a variety of touch receptors. They are **mechanoreceptors** that respond to different physical stimuli.

Touch receptor	Structure and location	Detects
Meissner's corpuscles	Endings of sensory neurons surrounded by a capsule; in the upper layer of skin (lips, fingertips, palm, soles)	Light touch and vibration
Hair follicle receptors	Endings of neurons around the base of hair follicles	Movement of hairs (mechanical or thermal stimuli)
Merkel cells	Oval-shaped receptors in the deepest layer of epidermis; in highly touch-sensitive areas like fingertips	Light touch
Pacinian corpuscles	Sensory neuron endings surrounded by a capsule; deeper in the skin	Deep pressure and vibration
Ruffini endings	Sensory neuron endings enclosed in a capsule; in the dermis	Sustained pressure and stretching

These receptors generate impulses and transmit them via sensory neurons to the spinal cord and then to the brain, where the signals are processed to create the sensation of touch.

Q.52 [Short] What are thermoreceptors? Where are they located?

Thermoreceptors are naked dendrite endings of sensory neurons. They detect fall and rise in temperature. Cold receptors are located immediately below the epidermis, while warm receptors lie slightly deeper, in the dermis. Thermoreceptors are also found in the hypothalamus, where they detect the temperature of the blood and monitor the core body temperature.

15.7 EFFECTS OF DRUGS ON NERVOUS COORDINATION

Q.53 [Short] Define drug, psychoactive drug and narcotic drug.

A **drug** is any substance (other than food) that is used to prevent, diagnose, treat or relieve the symptoms of a disease or abnormal condition.

A **psychoactive drug** alters brain functions and changes mood, perception, consciousness and behaviour, e.g., alcohol, caffeine, nicotine, cocaine and heroin.

A **narcotic drug** is a type of opioid drug that is used for pain relief and has a high potential for addiction and abuse, e.g., morphine, codeine and oxycodone.

Q.54 [Short] "All narcotics are psychoactive drugs, but all psychoactive drugs are not narcotics." Explain.

Narcotics alter brain functions, so every narcotic is a psychoactive drug. But many psychoactive drugs, such as caffeine, nicotine and alcohol, are not opioid pain-relieving drugs. So they are not narcotics.

Q.55 [Short] What is meant by dependence, drug addiction and drug tolerance?

Drug addiction is the disorder of uncontrollable drug desire and use despite its negative effects. The abuse of psychoactive drugs leads to dependence (the person relies on the drug physically or emotionally in order to function), and dependence often results in addiction.

Drug tolerance is the condition in which, as a person continues to use a drug, the body becomes adaptive and less sensitive to its effects. After tolerance, the person requires higher doses to get the same level of response. This leads to increased addiction.

Q.56 [Short] What are withdrawal symptoms?

When addicts stop taking an addictive drug, they experience withdrawal symptoms. These include anxiety, depression, fatigue, mood swings, tremors, headache, sweating, nausea, vomiting, loss of appetite, insomnia, breathing difficulties, rapid heart rate, high blood pressure, hallucinations and seizures.

studysapiens.com

Q.57 [Long] Explain the changes in the nervous system during addiction and tolerance.

1. Addictive drugs affect the regions of the brain which regulate pleasure and motivation. When these drugs reach the brain, presynaptic neurons release the neurotransmitter **dopamine**, which produces feelings of pleasure and reward.
2. The drug molecules block the dopamine reabsorption receptors in the presynaptic neuron. So a large amount of dopamine accumulates in the synaptic cleft.
3. The brain adapts and reduces the number of dopamine receptors on the postsynaptic neuron. This leads to **tolerance**, i.e., the user needs higher doses of the drug to get the same pleasure.
4. When the drug is stopped abruptly, dopamine release becomes normal, but due to fewer receptors the postsynaptic neuron is less stimulated. This leads to **withdrawal symptoms**.

Q.58 [Short] Explain addiction and tolerance of caffeine.

Caffeine is found in many common beverages, such as coffee, tea and energy drinks. It blocks the action of a neurotransmitter called **adenosine**, which promotes sleep and relaxation. As a result, caffeine increases alertness and energy levels. Regular use of caffeine leads to **tolerance**, so the user requires higher doses to achieve the same level of alertness. If caffeine use is stopped abruptly, withdrawal symptoms such as headaches and fatigue occur.

Q.59 [Short] Explain addiction and tolerance of nicotine.

Nicotine is a highly addictive stimulant present in tobacco products (cigarettes, cigars and smokeless tobacco). It binds to nicotine receptors in the brain. These receptors release dopamine and other neurotransmitters, which produce pleasurable feelings. Regular use leads to **tolerance**, so the user requires higher doses to achieve the same pleasure. When nicotine is stopped abruptly, the user experiences withdrawal symptoms such as irritability, anxiety and cravings. Long-term use of nicotine is also associated with an increased risk of cancer, heart disease and respiratory problems.

Q.60 [Long] Differentiate between use and abuse of drugs. Describe the abuse of common psychoactive drugs.

“Use” refers to the appropriate or prescribed use of drugs for medical purposes. “Abuse” generally refers to the inappropriate use of psychoactive drugs.

Drug	Source / form	Effect on user	Negative effects
Heroin	Highly addictive opioid from the opium poppy plant; injected, smoked or snorted	False sense of euphoria and relaxation; quickly leads to dependence and addiction	Respiratory depression, infections, mental health disorders

Drug	Source / form	Effect on user	Negative effects
Cannabis	From dried flowering tops, leaves and stem of Indian hemp (*Cannabis sativa*); marijuana, hashish	Relaxation, euphoria, altered perception, increased appetite	Impaired cognitive function, respiratory problems, mental health disorders
Nicotine	Tobacco products: cigarettes, cigars, smokeless tobacco	Addiction and dependence	Respiratory problems, cancers, cardiovascular disease, mental health disorders
Alcohol	Beer, wine and spirits	Relaxation and euphoria	Impaired cognitive and motor functions, liver diseases, cardiovascular diseases, mental health disorders
Inhalants	Nail polish remover, glue (inhaled)	Euphoria, relaxation, altered perception	Respiratory problems, cognitive impairment, neurological damage

15.8 DISORDERS OF NERVOUS SYSTEM

Q.61 [Short] How are nervous disorders classified?

They are classified into five groups: **vascular** (e.g., stroke), **infectious** (e.g., meningitis), **structural** (e.g., brain tumour, spinal cord injuries), **functional** (e.g., headache) and **degenerative** disorders (e.g., Alzheimer's disease).

Q.62 [Long] What is a stroke? Give its causes, symptoms and treatment.

A **stroke** means damage to brain cells due to an interruption in the blood supply to a part of the brain. The interruption may be due to a blockage or a rupture of a blood vessel of the brain.

	Stroke
Causes	High blood pressure, smoking, diabetes, high cholesterol, obesity and family history of stroke.
Symptoms	They vary with the area of brain affected: paralysis on the side of the body opposite to the damaged area; aphasia (inability to express through words); sudden trouble in seeing and walking; dizziness, loss of balance; severe headache.
Treatment	Given as soon as possible: medications (blood thinners, clot-dissolving drugs, antihypertensive drugs); surgery to remove a blood clot or repair a ruptured blood vessel; rehabilitation to recover physical and cognitive function.

Q.63 [Short] Differentiate between ischemic stroke, haemorrhagic stroke and transient ischemic attack.

Ischemic stroke occurs when the blood vessels of the brain become narrowed or blocked. **Haemorrhagic stroke** occurs when a blood vessel in the brain leaks or ruptures. **Transient ischemic attack** (ministroke) is caused by a temporary decrease in blood supply to a part of the brain, which may last as little as five minutes.

Q.64 [Short] What is meningitis? Give its causes, symptoms and treatment.

Meningitis is an inflammation of the meninges due to infection.

	Meningitis
Causes	Bacterial meningitis (most severe and life-threatening); viral meningitis (usually less severe); fungal meningitis (rare, usually in people with weak immune systems); parasitic infections.
Symptoms	Stiffness in the neck, headache and fever. In severe cases, paralysis or coma.
Treatment	Bacterial: immediate treatment with antibiotics. Viral: usually self-limiting; antiviral medications may be used. Fungal: long-term antifungal medications. Vaccines can prevent some bacterial meningitis.

Q.65 [Short] What is a brain tumour? Give its causes, symptoms and treatment.

A **brain tumour** is an abnormal growth or mass of cells in the brain. It can be benign (non-cancerous) or malignant (cancerous). It can arise from glial cells or from the neurons themselves.

	Brain tumour
Causes	The exact cause is unknown. Risk factors include exposure to radiation, family history and certain genetic mutations.
Symptoms	They depend on the location of the tumour: headaches, severe nerve pain, paralysis, seizures and coma.
Treatment	Surgery, radiation therapy and chemotherapy.

Q.66 [Short] What are functional disorders? Write a note on headache.

Functional disorders affect the normal functioning of the nervous system without any apparent structural or organic cause. **Headache** is pain or discomfort in the head or neck. **Primary headaches** are due to the headache itself, e.g., migraine and tension headache. **Secondary headaches** are due to an underlying structural problem such as bleeding in the brain, tumour or meningitis. Treatment includes pain-killing (analgesic) medicines, reducing stress, regular exercise and better sleep habits.

Q.67 [Short] What is migraine?

Migraine is a type of headache with intense, throbbing pain on one side or in one region of the head. It is caused by changes in the brain and blood vessels, which lead to inflammation and pain.

Q.68 [Short] What are degenerative disorders? Give examples.

These are conditions in which there is progressive deterioration of nerve cells in the brain, spinal cord or peripheral nerves.

- **Parkinson's disease:** progressive death or weakening of the brain neurons that produce dopamine; causes movement issues and cognitive changes.
- **Huntington's disease:** progressive damage to neurons in the brain; causes movement disorders, cognitive decline and psychiatric symptoms.
- **Multiple sclerosis:** autoimmune disease of the CNS in which the myelin sheath is damaged; causes vision loss, pain, fatigue and impaired coordination.
- **Alzheimer's disease.**

Q.69 [Short] What is Alzheimer's disease? Give its causes, symptoms and treatment.

Alzheimer's disease is a progressive degenerative brain disorder that affects memory, thinking and behaviour. It leads to dementia in older adults. Its onset usually occurs in aged people, but it is not particularly associated with aging.

	Alzheimer's disease
Causes	A combination of genetic, environmental and lifestyle factors.
Symptoms	It begins with mild memory loss and confusion and progresses to more severe cognitive impairment, including difficulty with language, problem-solving and completing familiar tasks.
Treatment	There is no cure, but there are medicines to slow its progression and to improve memory and cognitive function.

15.9 DIAGNOSTIC TESTS FOR NERVOUS DISORDERS

Q.70 [Short] Explain the principle, procedure and uses of EEG.

Principle: EEG measures the electrical activity of the brain. It is based on detecting the electrical nerve impulses being transmitted among the neurons.

Procedure: Metal electrodes are attached to the scalp and connected to an amplifier. The amplifier magnifies the electrical signals detected by the electrodes. The signals are recorded and displayed as a series of waveforms on a computer screen or paper.

Uses: EEG is done to evaluate neurological conditions, including seizures, epilepsy and brain tumours. It is painless and does not require sedation or anaesthesia.

Q.71 [Short] Explain the principle, procedure and uses of CT scan.

Principle: CT scan uses X-rays and computer processing to generate detailed images of the body's internal structures. It is based on the fact that different tissues of the body absorb X-rays to different degrees.

Procedure: The patient lies on a motorized table that slides into a doughnut-shaped machine called a gantry. The gantry contains an X-ray tube that rotates around the patient. Multiple detectors on the opposite side record the absorption of X-rays through different tissues. A computer uses this information to construct a detailed, cross-sectional image.

Uses: CT scans are useful for imaging soft tissues such as the brain, lungs and abdomen, and for detecting abnormalities in bone structure. They are used to diagnose cancers, infections and injuries.

Q.72 [Short] Explain the principle, procedure and uses of MRI.

Principle: MRI uses a strong magnetic field and radio waves to produce detailed images of the body's internal structures. It is based on the fact that different tissues have different amounts of hydrogen atoms, which have magnetic properties.

Procedure: The patient lies on a motorized table that slides into a large tube-like machine called a scanner. Its powerful magnet creates a strong magnetic field around the body. Radio waves then cause the hydrogen atoms to produce signals, which are detected by a receiver. A computer uses this information to construct a detailed, three-dimensional image.

Uses: MRI is used for imaging soft tissues such as the brain, spinal cord and internal organs, and for detecting abnormalities in blood vessels and certain types of tumours.

Q.73 [Short] Compare CT scan and MRI.

CT scan	MRI
Uses X-rays and computer processing.	Uses a strong magnetic field and radio waves.
Based on different absorption of X-rays by different tissues.	Based on different amounts of hydrogen atoms in different tissues.
The machine is a doughnut-shaped gantry.	The machine is a large tube-like scanner.
Gives cross-sectional images.	Gives three-dimensional images.